

Hadoop

Database

Datalake

Hadoop = H-DFS + MapReduce

FS distributed on multiple machines

/ - root dir

f1.txt - 1MB - blocks 100KB

Vertical scalability

1 TB + 1 TB

32 cores - 64 cores

64 GB - 128 gb

select * from emp - 30 sec

1 GB / 128 MB = 8 blocks

b1(b11) b2

b3 b4

b5 b6

b7 b8

M - 1 TB, 16GB, 16 cores

4 TB, 64GB 64 cores

64 mb

M1 - b1 b2 b41 b51
M2 - b3 b4. b11 b81
M3 - b5 b6. b21 b71
M4 - b7 b8 b31 b61
m5-8

Parallel Processing - 1 task for 1 block, 8 tasks, 1 task 2 second

file.txt
1gb file. 1 min

```
f = open('file.txt')
d = dict()
for line in f.readlines():
    words = line.split(" ")
    for w in words:
        if w in d:
            d[w] = d[w]+1
        else:
            d[w] = 1
```

128 MB

M1 - b1- mapper task T1 b2 -T2
M2 - b3 b4.
M3 - b5 b6.
M4 - b7 b8

<0,hello world hi world> - b1 - {hello:1, hi:1, world:2}
hi world hello asia - b2 - {hi:1, world:1 hello:1 asia:1}

hello asia -b3 - {hello:1, asia:1}
hi asia - b4 {hi:1, asia:1}
hello nepal - b5 {hello:1 nepal:1}
hi india - b6. {hi:1, india:1}
hadoop hdfs - b7 {hadoop:1 hdfs:1}
java python -b8 {java:1 python:1}

{hello:4, hi:4, world:3, asia: 3, nepal:1, india:1,
hadoop:1, hdfs:1, java:1, python:1}

<k,v> Mapper <k,v>
<k,v>Reduce <k,v>
hascode() -
hello - 10 % 2
hi - 9 % 2
world - 11 % 2
asia -8 % 2
india - 5 %2
nepal - 7
hadoop - 20
python - 13
java - 15

R0 -[hello:1, hello:1,hello:1,hello:1, asia:1,asia:1,asia:1]
- hello : 4, asia:3
R1 - [hi:1, hi:1,hi:1,hi:1, world:2, world:1, india:1] - hi:3,
word:3, india:1

select word,count(*) from table group by word

```
job = fileInput('file.txt')
```

```
select name, sum(salary) from emp group name
```